WASTEWISE SCAN · REGULATORY AND STANDARDS REGISTER

WasteWise Scan regulatory and standards register

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-24	Amish Chadha	First regulatory and standards register (kit 1.8 upgrade), for a first pilot with a waste picker cooperative in Pune, India; India waste rules cited through the focus-area register of WasteWise-ml (WML-REG-001)

WasteWise Scan regulatory and standards register

Which standards and regulations apply, where, and what is needed, for a first pilot in Pune, India (first candidate partner SWaCH, decided 2026-10-02). The scanner is a battery-powered handheld with invisible infrared LEDs and a radio-capable controller board; it advises on resin and never certifies material or judges whether waste is safe. The items that need work before any product claim are eye safety (IEC 62471, met by estimate only), the lithium-ion cell (BIS registration of cells to IS 16046 Part 2), radio approval of the controller board in India (WPC), and battery and e-waste producer duties if units are ever supplied. Standards shared with WasteWise-ml (the Circular Materials focus area) are cited from its register, WML-REG-001. Not legal advice; anything marked "to assess" needs a qualified reviewer before any product claim. Method: STANDARDS section 23.

Table 1. Regulatory and standards register.

#	Standard or regulation	Jurisdiction	What it covers	Applicability to this design	Owner (role)	Status
R1	IEC 62471:2006 (EN 62471), photobiological safety of lamps and lamp systems	International	Risk groups (exempt, 1 to 3) for optical radiation from lamps and LEDs, including infrared to the eye	Eight NIR and SWIR LEDs pulsed about 14 ms each per scan; PR-014 targets the exempt group	Design lead	Met by design on paper (WSC-CAL-001 H: 3.9 W/m2 at 200 mm against 100 W/m2); limits to confirm against the purchased standard and LED datasheets; measurement at TRL 4
R2	Hardware limit on LED on-time (practice that keeps R1 true under a firmware fault)	Design practice	A stuck-on LED at the shroud rim could exceed the corneal limit over long exposures	Required by PR-014 and WSC-PRC-001 Safety	Electronics lead	Applies; to design at TRL 4 (monostable or RC limit on each driver)
R3	IS 16046 (Part 2): 2018, equivalent to IEC 62133-2:2017, safety of portable sealed secondary lithium cells and batteries; BIS Compulsory Registration Scheme	India	Cell and battery safety tests; registration of cells sold in India	The protected 18650 cell (BOM line 8)	Procurement (partner)	To assess: buy a cell registered with BIS under this standard (PR-015); which cell categories the scheme lists is to confirm on the BIS site
R4	UN Manual of Tests and Criteria, section 38.3, transport of lithium cells	International	Tests before lithium cells or equipment containing them are shipped	Shipping cells or assembled scanners	Procurement	To assess (not checked in this session): buy cells with a UN 38.3 test summary
R5	Wireless Planning and Coordination (WPC) Equipment Type Approval for equipment in delicensed bands	India	Devices using 2.4 GHz Wi-Fi or Bluetooth need ETA; an end product made in India needs no separate ETA if its RF module already has one and is fitted unmodified (Granite River Labs guide)	The ESP32-S3 display board has a 2.4 GHz radio; the firmware keeps it off, but the module is radio equipment	Procurement (partner)	To assess: buy a board whose module holds an ETA certificate, or confirm the radio-off case with a qualified reviewer
R6	Battery Waste Management Rules, 2022 (notified 2022-08-24)	India	Extended producer responsibility for all battery types including portable; collection and	Only if scanners with cells are supplied in India; each used cell goes to a	Project lead; partner	To assess before any supply; applies to the partner's disposal of used cells now

#	Standard or regulation	Jurisdiction	What it covers	Applicability to this design	Owner (role)	Status
			recycling, no landfill or incineration (pv magazine India)	battery collection route		
R7	E-Waste (Management) Rules, 2022	India	Electrical and electronic equipment at end of life	The scanner itself at end of life	Project lead; partner	To assess (cited through WML-REG-001 G5; not checked in this session)
R8	Solid Waste Management Rules, 2026	India	Segregation at source and handling of dry recyclables; special care waste	The site where the scanner is used; the scanner never decides whether an item is special care waste	Partner (site operator)	Applies to the site (cited through WML-REG-001 G1)
R9	ASTM D7611/ D7611M-21, resin identification coding	International	The resin codes moulded on plastic items	The codes the scanner displays (PR-001, PR-007)	Design lead	Applies (reference)
R10	IEC 60529, degrees of protection by enclosures (IP code)	International	IP54 target of PR-010	Shell gasket, sealed button, window	Design lead	To verify at TRL 4 (spray test); the standard is cited by number only, not purchased
R11	Digital Personal Data Protection Act, 2023, and Rules, 2025	India	Personal data in digital form	Scan records hold no personal data by design; a pair ID joins to WasteWise-ml photos, which may	Data lead with the partner	Does not apply to the scanner's own record by design; joint records to assess (cited through WML-REG-001 G2)
R12	RoHS-type substance restrictions (India E-Waste Rules carry RoHS limits)	India	Hazardous substances in electrical equipment	Bought boards and LEDs	Procurement	To assess (not checked in this session)
R13	Medical, food-contact or hazardous-material claims	Any	Certifying material grade or safety	Out of scope: the scanner gives sorting advice only (WSC-PRB-001, Out of scope; WSC-PRC-001 Safety)	Project lead	Does not apply by design; keep the claim out of all user material

Status: to assess, applies, does not apply, met by design, certification needed later.

Sources

- BAuA. *Photobiological Safety of Light Emitting Diodes (LEDs)*. Report F 2115, 2013 (EN 62471 exempt group and risk groups 1 to 3). https://baua.de/EN/Service/Publications/Report/F2115. Checked 2026-10-24.
- Bureau of Indian Standards. IS 16046 (Part 2):2018 test report format (title: safety requirements for portable sealed secondary cells, part 2 lithium systems; equivalent to IEC 62133-2:2017). https://www.bis.gov.in/wp-content/uploads/2026/03/UTRF_16046_2_2018.pdf. Checked 2026-10-24.
- Granite River Labs. "A Guide to WPC and ETA Certification." https://www.graniteriverlabs.com/en-us/market-access-services/country/india/wpc-mark. Checked 2026-10-24.
- pv magazine India. "Government notifies Battery Waste Management Rules, 2022." 2022-08-25. https://www.pv-magazine-india.com/?p=10815. Checked 2026-10-24.
- WasteWise-ml regulatory and standards register WML-REG-001 v0.1 (Solid Waste Management Rules, 2026; DPDP Act and Rules; E-Waste Rules), the Circular Materials focus-area register.
- ASTM D7611/D7611M-21, as cited in WSC-PRB-001. https://store.astm.org/d7611_d7611m-21.html.